CAPSTONE PROJECT REPORT

Bring Your Own Project (BYOP) | Submission File

SUBJECT: FUNDAMENTALS OF AI & ML

NAME: DEVANSH CHOUDHARY

REGISTRATION NO.: 25BCE10575

1. The Problem Chosen

In the modern digital economy, individuals frequently make multiple transactions across various platforms (UPI, ATMs, Net Banking). However, keeping track of these micro-transactions and understanding overall spending habits remains a significant challenge. Traditional banking apps often provide raw, overwhelming lists of transactions without offering immediate, easy-to-understand visual trends of a user's financial health. The problem I chose to solve is the lack of a simple, localized, and visually intuitive tool for personal financial tracking.

2. Why It Matters

Financial awareness is a critical life skill. When people cannot easily visualize their cash flow (deposits vs. withdrawals), they are more prone to overspending and poor financial planning. This project matters because it bridges the gap between raw financial data and actionable insights. By providing a tool that not only logs transactions but instantly generates visual reports, users can effortlessly identify their spending patterns and manage their savings more effectively.

3. Approach to Solving It

I approached this problem by building a robust, console-based Python application utilizing the Object-Oriented Programming (OOP) paradigm.

- **Core Logic & State Management:** I designed a FinancialAssistant class to encapsulate all transaction data and core functionalities. This keeps the global namespace clean and the code modular.

- **Data Storage:** I used Python lists to store transaction histories as tuples (Category, Amount, Timestamp, Location). For the monthly aggregation, I utilized the collections.defaultdict, which provides a highly efficient way to group transactions by month.
- **Data Visualization:** To solve the visualization problem, I integrated the matplotlib.pyplot library. This transforms basic terminal output into dynamic bar charts and line graphs.
- **User Interface:** I implemented a while loop-based Command Line Interface (CLI) that continuously presents a menu until the user explicitly chooses to exit, ensuring a seamless user experience.

4. Key Decisions Made

Throughout the development process, several architectural decisions were made to enhance functionality and user experience:

- **Simulating Real-World Data:** To make the application feel authentic, I implemented a random location generator (e.g., "ATM - Bhopal", "UPI Payment") using the random module. This mimics actual bank statements.
- **Robust Input Validation:** I recognized that users might accidentally input text instead of numbers. To prevent the program from crashing, I implemented try-except blocks handling ValueError exceptions during deposits and withdrawals.
- **Modular Visualization:** Instead of cramming plotting logic into the main loop, I created dedicated methods (plot_history and plot_monthly_report). This adheres to the Single Responsibility Principle, making the code easier to maintain and read.

5. Challenges Faced

- **Time-Series Data Aggregation:** One significant challenge was grouping individual transactions by month for the Monthly Report. I solved this by slicing the timestamp string (timestamp[:7] to extract YYYY-MM) and dynamically updating a defaultdict to calculate net savings per month.
- **Handling Empty States:** Initially, requesting a chart before any transactions were made would cause matplotlib to throw an error. I resolved this by adding guard clauses (if not self.history: return) to gracefully notify the user that no data is available to plot yet.
- **Data Synchronization:** Ensuring the balance variable in the main() function stayed perfectly synced with the FinancialAssistant history list required careful tracking of the deposit() and withdraw() return values.

6. What I Learned

This capstone project was an excellent synthesis of everything covered in the Python Essentials course. Key takeaways include:

- **Library Integration:** I gained practical experience installing and utilizing third-party libraries (matplotlib) alongside standard built-in modules (datetime, collections).
- **Advanced Data Structures:** I learned how powerful tools like defaultdict can be for sorting and aggregating complex datasets compared to standard dictionaries.
- **Defensive Programming:** Implementing comprehensive try-except blocks taught me how to write resilient code that expects and handles user errors gracefully.
- **Software Design:** Structuring the program with functions and classes made debugging

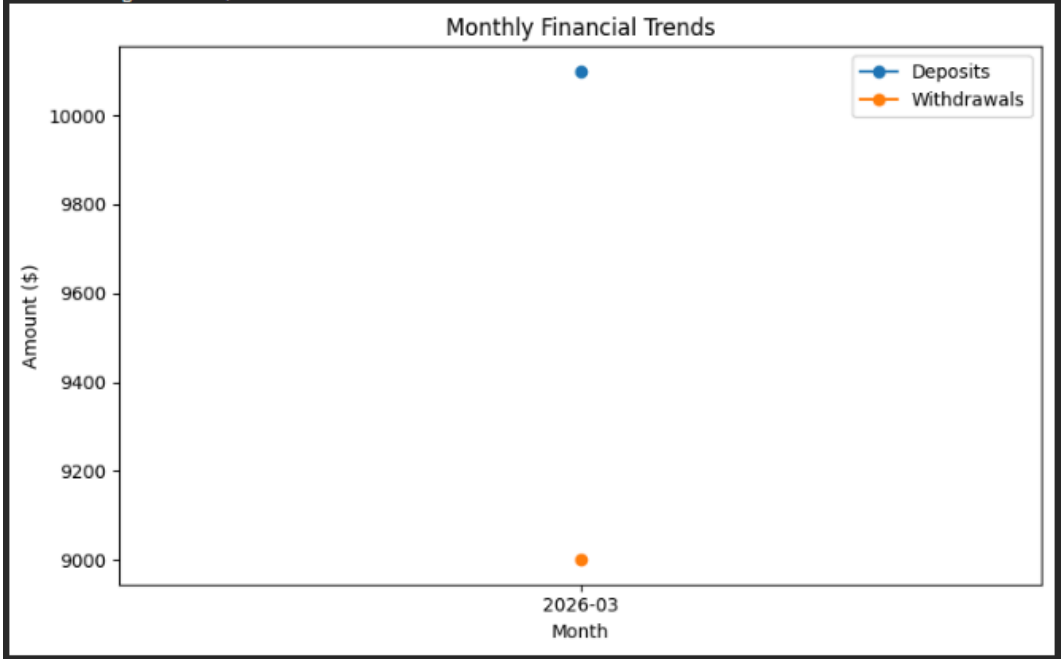
significantly easier and prepared me for larger-scale software development.

7. Expected Program Output

Below is a simulated run of the application, demonstrating the menu interface, transaction handling, and analytical readouts:

```
***********************
   BANKING MENU
***********************
1. Show Balance
2. Deposit Money
3. Withdraw Money
4. View AI Analytics & Chart
5. View Monthly Report
6. Exit

Your Current Balance is: $10000.00

***********************
   BANKING MENU
***********************
1. Show Balance
2. Deposit Money
3. Withdraw Money
4. View AI Analytics & Chart
5. View Monthly Report
6. Exit

Your Current Balance is: $7000.00

***********************
   BANKING MENU
***********************
1. Show Balance
2. Deposit Money
3. Withdraw Money
4. View AI Analytics & Chart
5. View Monthly Report
6. Exit

Your Current Balance is: $7100.00

***********************
   BANKING MENU
***********************
1. Show Balance
2. Deposit Money
3. Withdraw Money
4. View AI Analytics & Chart
5. View Monthly Report
6. Exit

Your Current Balance is: $1100.00

***********************
   BANKING MENU
***********************
1. Show Balance
2. Deposit Money
3. Withdraw Money
4. View AI Analytics & Chart
5. View Monthly Report
6. Exit
```

```
**********************
   BANKING MENU
**********************
1. Show Balance
2. Deposit Money
3. Withdraw Money
4. View AI Analytics & Chart
5. View Monthly Report
6. Exit

--- AI Monthly Financial Report ---

Month: 2026-03
  Total Deposits:    $10100.00
  Total Withdrawals: $9000.00
  Net Savings:       $1100.00
```




```
**********************
   BANKING MENU
**********************
1. Show Balance
2. Deposit Money
3. Withdraw Money
4. View AI Analytics & Chart
5. View Monthly Report
6. Exit
```

```
--- AI Financial Summary ---
Total Deposited:   $10100.00
Total Withdrawn:   $9000.00
Net Cash Flow:     $1100.00
Transaction Count: 4

--- Detailed Transaction History ---

1. Deposit of $10000.00
   Time: 2026-03-28 04:11:46
   Location: UPI Payment

2. Withdrawal of $3000.00
   Time: 2026-03-28 04:12:08
   Location: Mobile App

3. Deposit of $100.00
   Time: 2026-03-28 04:12:17
   Location: Mobile App

4. Withdrawal of $6000.00
   Time: 2026-03-28 04:12:25
   Location: Bank Branch
```

```
**********************
   BANKING MENU
**********************
1. Show Balance
2. Deposit Money
3. Withdraw Money
4. View AI Analytics & Chart
5. View Monthly Report
6. Exit
Selection: 6
Thanks for visiting!
```